

Fanglu (Louie) Yang

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EDUCATION

University of Toronto

Sep 2023 – Present

HBSc – Co-op Specialist in Statistics – Statistical Machine Learning and Data Science Stream

Major in Computer Science, Minor in Economics

Awards: E-Fund Entrance Scholarship of Excellence - 10,000 CAD, 2024 Dean's list

WORK EXPERIENCE

Jiangsu X-chip Semiconductor Technology Co., Ltd.

June 2026 – Dec 2026

AI Automation Engineer Intern – **vLLM, RAG, Fine-tuning, Embedded C, Docker**

Shanghai, China

- Architected an enterprise-grade, air-gapped AI automation platform from scratch utilizing **Docker and vLLM** on an RTX Pro 6000, deploying quantized 32B/72B models with dynamic KV Cache monitoring to slash inference latency by 45% without OOM faults.
- **Fine-tuned** local LLMs via **LoRA** on domain-specific ATE instructions and register-mapping datasets, tailoring model weights for embedded C driver synthesis and test script generation while cutting test development overhead by **80%**.
- Engineered an automated ATE verification pipeline pairing multi-modal **VLMs** and pattern recognition to parse oscilloscope screenshots, routing structured signal metrics into an LLM reporting engine to generate batch Excel/PPT validation decks and slashing manual review time by **90%**.

Baker Tilly Canada

May 2025 – Aug 2025

Data Science Co-op – **LLM, LangChain, Embedding, Streamlit, Python, Alteryx, Power BI**

Toronto, Canada

- Designed and deployed a **SOC 2 audit automation system** with interactive ML-driven control testing dashboard and **LLM chatbot** for policy search and evidence interpretation, cutting manual workload by **90%**.
- Developed an end-to-end **ML pipeline (LangChain, Gemini, Python)** that uses LLM-driven judgment to assess evidence across **80+** controls, increasing audit flagging accuracy by **85%** and reducing review time by **70%**.
- Implemented **RAG with vector search and embeddings (LangChain, FAISS)** to power dynamic SOC testing queries and policy lookups, enhancing interpretability and decision accuracy by **85%**.
- Engineered **recursive OCR + anonymization system (Tesseract, SpaCy)** for text extraction and entity masking and integrated Gemini to automate compliance decisions, cutting document processing time by **75%**.
- Designed **Alteryx workflows** for automated data cleaning, merging, and transformation and performed interactive **Power BI dashboards**, streamlining analysis speed for audit GL review and tax compliance reporting by **80%**.

PROJECTS

TTC Delay Analytics and Prediction - Datathon: 1st Winner (Best Overall)

Mar 2025

UoT SDSS (Students in Data Science and Statistics) Datathon

- Evaluated **10 datasets(150k+ entries)** of Subway, Streetcar and Bus delay data using Python (Pandas, Numpy) and achieved **over 99% accuracy** on lateness prediction with **Correlation Heatmap, Logistic Regression, Classification, and XGBoost**.
- Collaborated **analytical analysis** with Data Scientists and UI/UX Designer to integrate **PowerBI** and **Figma** for interactive dashboard design with **50+ dynamic filtering**, gauge charts, data tables, bar and line graphs.
- Trained an **XGBoost classifier** with **98.92% accuracy** using hyperparameter tuning via **GridSearchCV**.
- Won **Best Overall Project** out of **100 students** in consideration of Visualization, Modeling, and Business Insights.

Statistical Analysis on Research Project Investments by the MCU

Mar 2024 – Apr 2024

Analyzed a 20-year (2004–2024) dataset from the Ministry of Colleges and Universities (Coursework Project)

- Implemented **RStudio** for data cleaning, wrangling, statistical analysis (bootstrap, clustering, cross-validation, logistic regression, random forest), and data visualization (ggplot) on a dataset with **5000+ entries**.
- Conducted **A/B testing** from different approaches, recommending Govt of Ontario to diversify research investment.

SKILLS

Languages: Python(Pandas, NumPy, Matplotlib, Scikit-learn, Seaborn, Keras, PySpark, XGBoost, GridSearchCV, Tensorflow, spaCy), C, Java, JavaScript, R, SQL, HTML/CSS

AI, ML & Data Science: Model Fine-Tuning (LoRA), Local LLM/VLM (vLLM, Docker), RAG, Agentic Workflows, Computer Vision, NLP, Scikit-learn, XGBoost, PySpark, Data Pipeline Design

Tools & Frameworks: Git, PySide6, SQLite, Power BI, Alteryx, Azure, PostgreSQL, Linux, Embedded HAL Design